

Command	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
Issue						A separate command issuer chooses stored commands, sends them in order, repeats sequences, and waits for required results. It sends work to both the SIMD unit (parallel lane operations) and the matrix unit. The SIMD unit reports completion or an error; it does not fetch a program or branch by itself. LUT means lookup table; here it stores complete command words.																															
Mask						A lane is one parallel processing position. VL is the number of positions currently in use, starting at lane 0. PM selects a predicate register whose true bits allow those positions to execute. A lane executes only when its index is below VL and its selected predicate bit is true. Other lanes keep their previous values. A written result fills its low 16/32/64 bits; the remaining bits of the 64-bit register cell become zero.																															
State						The command issuer manages the execution context: vector registers V, scalar registers S, predicate registers P, active length VL, and diagnostic counters. It waits for earlier writes before issuing commands that read them. Separate memory logic moves values between memory and registers and fetches entries from data lookup tables.																															
Unused						A field that an operation does not use must contain zero, rather than an arbitrary register number. FAMILY=111 and any FAMILY/SIG combination absent from Signatures cause an error before any result is written.																															

[1] CAST type numbers: 0 F16, 1 F32, 2 I8, 3 U8, 4 I16, 5 U16, 6 I32, 7 I64. $SIG = FROM \times 8 + TO$; SIG bit 6 is zero. Only the 24 pairs in Signatures are valid. U32/U64 storage and arithmetic remain supported, but are not CAST endpoints.